

Fermionic Matrix Models, Fortuity, and SYK

Abstract. We study a three-parameter family of purely fermionic quantum mechanics: p flavours of adjoint matter for a gauge algebra of rank r , with a nilpotent supercharge of odd degree q built from traces. The family contains $\mathcal{N} = 2$ SYK as its rank-one member, since at $N = 1$ the trace is the identity map and Q becomes a generic q -fermion supercharge on p modes. The BPS states are the cohomology of multiplication by a fixed q -form, and the width of the R-charge window they occupy is set by the rank, $W = r(w_s - 1) + 1$ at the maximal weight, where w_s is the width contributed by a single Cartan site. We then solve for w_s . For a generic q -form each grading class with non-vanishing Witten index carries cohomology in exactly one degree, of dimension $|E_c|$, so index saturation and R-charge concentration are the same statement for the rank-one model. Adjoining a flavour realises the complex as a mapping cone, and the whole of w_s is controlled by one induced map δ whose source and target are Poincaré dual. It is therefore a bilinear form, symmetric or antisymmetric according to the parity of $j = \frac{1}{2}(p - q + 1)$; a generic symmetric form is non-degenerate while an antisymmetric one on an odd-dimensional space cannot be, and this alternation makes w_s cycle through $q - 1, q, q + 1, q$ with period four in p . Consequently $W = r(q - 2) + 1$ at best, concentration needs $r \leq (q - 1)/(q - 2)$, and only $q = 3$ tolerates a gauge group, at rank 2. The mechanism behind the rank law is the Weyl group: at the maximal weight the complex is the Cartan sector carrying a q -form that gauge invariance forces to be Weyl invariant, and while generic q -forms have bounded window, Weyl invariant ones do not. Holding the Cartan at twelve modes and varying only the Weyl group gives $W = 3, 3, 5, 5, 11$ for $|\mathcal{W}| = 1, 2, 6, 24, 720$. If -1 lies in the Weyl group, as for B, C, D_{even} and most exceptional algebras, the Cartan form vanishes and the window is maximal, so the unitary series is already optimal. Fortuity runs the opposite way and is total: an embedded BPS state is never closed at the next rank.

1 One family

Let $\mathfrak{g}$ be a reductive Lie algebra of rank r with a fixed invariant form, and let Ψ^a , $a = 1, \dots, p$, be complex fermions valued in $\mathfrak{g}$. For $\mathfrak{g} = \mathfrak{gl}(N)$ we write Ψ_{ij}^a with $i, j = 1, \dots, N$ and $\{\Psi_{ij}^a, \bar{\Psi}_{kl}^b\} = \delta^{ab} \delta_{il} \delta_{jk}$, the gauge group acting by conjugation. The Fock space is the exterior algebra

$$\mathcal{H} = \Lambda^\bullet V, \quad V = \mathbb{C}^p \otimes \mathfrak{g}, \quad n \equiv \dim V = p \dim \mathfrak{g}, \quad (1)$$

graded by the R-charge $k = N_\Psi = \text{Tr} \Psi^a \bar{\Psi}^a$, so $\Lambda^k V$ is the k -fermion sector. A basis vector of V is a *mode*; a basis vector of $\Lambda^k V$ is a set S of k modes, written m_S . All dynamics comes from one supercharge,

$$Q = \sum_{1 \leq a_1 < \dots < a_q \leq p} C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}], \quad H = \{Q, \bar{Q}\} \geq 0, \quad [N_\Psi, Q] = qQ. \quad (2)$$

Only odd q is admissible, for two independent reasons: a cyclic shift inside $\text{Tr}[\Psi^q]$ costs $(-1)^{q-1}$, so the trace vanishes identically for even q ; and $Q^2 = 0$ needs the exchange of the two q -tuples in $C \otimes C$, a permutation of $2q$ fermions built from q^2 transpositions, to produce a relative sign, which again happens exactly for odd q .

Since every Ψ is a creation operator, Q contains no annihilators and acts on $\Lambda^\bullet V$ as plain left multiplication,

$$Q = \omega \wedge -, \quad \omega = \sum_{1 \leq a_1 < \dots < a_q \leq p} C_{a_1 \dots a_q} \text{Tr}[\Psi^{a_1} \dots \Psi^{a_q}] \in \Lambda^q V, \quad H^\bullet = \text{Ann}(\omega)/(\omega). \quad (3)$$

The whole problem is the cohomology of multiplication by one specific element of $\Lambda^q V$. Note also that $\omega \wedge \omega = 0$ automatically for odd q , so nilpotency survives any modification of the couplings or of the index structure, a fact we use freely in §4.

SYK is already inside this family rather than a limit of it. At $N = 1$ the matrices are 1×1 , the trace is the identity map, and (2) reads

$$Q = \sum_{1 \leq a_1 < \dots < a_q \leq p} C_{a_1 \dots a_q} \psi^{a_1} \dots \psi^{a_q}, \quad (4)$$

which is $\mathcal{N} = 2$ SYK with p complex fermions. The gauge algebra is $u(1)$, whose adjoint action is trivial, so nothing constrains C ; this is why the SYK coupling is free while the matrix coupling is not. Randomness plays no role in what follows. What a Gaussian ensemble supplies is genericity, and genericity is optimal here for a structural reason: matrix ranks are lower semicontinuous in ω , so specialising the couplings can only enlarge the

(a) maximal weight freezes the off-diagonal

	$j=1$	$j=2$	$j=3$
$i=1$	free	p	p
$i=2$	0	free	p
$i=3$	0	0	free

occupation n_{ij} at λ^*

(b) only self-loops can close a cycle

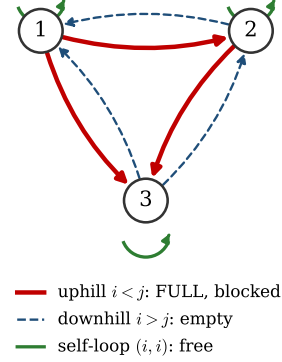


Figure 1: (a) At the maximal weight the occupation is forced: every mode with $i < j$ is full, every mode with $i > j$ is empty, and only the diagonal is free. (b) Reading modes as coloured directed edges, Q creates a closed walk $i \rightarrow j \rightarrow k \rightarrow i$ and needs all its edges empty. Uphill edges are occupied, so every creatable step is non-increasing; a cycle of non-increasing steps must be constant, leaving only self-loops.

cohomology. We checked the identification rather than assuming it, by running the full matrix machinery at $N = 1$ against a bare generic q -form; the two agree for every p from 4 to 12, with identical Betti numbers. What varies along the family is therefore not the coupling tensor but the *interaction hypergraph*. Calling a *triangle* the support of a non-zero term of ω , SYK has all $\binom{p}{q}$ of them, whereas at general N only the index-loop triangles survive out of the $\binom{pN^2}{q}$ available, a fraction of order $6/N^3$ at $q = 3$. The parameter N dials the hypergraph from complete to sparse.

The Cartan torus acts on Ψ_{ij} by $u_i u_j^{-1}$, so that mode carries weight $e_i - e_j$ and a state's weight is the row sums minus the column sums of its occupation matrix; diagonal modes carry weight zero. Since Q is a gauge singlet it preserves the weight, and since $[N_\Psi, Q] = qQ$ it raises k by q , so the cohomology splits into complexes labelled by $(\lambda, \omega_\text{fl}, c)$ with λ a $U(N)$ irrep, $\omega_\text{fl} \in \mathbb{Z}_p$ a flavour charge, and $c = k \bmod q$. R-charge concentration, in the sense of Chang, Chen, Sia and Yang, says that in each complex of macroscopic index all BPS states sit at one degree. Writing W for the number of consecutive R-charges carrying BPS states, concentration therefore requires $W \leq q$, since a window wider than the grading period must place two occupied degrees in one residue class. Everything below determines W .

Both ends of the family have been studied. At $N = 1$ concentration was formulated by Chang, Chen, Sia and Yang [2], whose §5.1 shows empirically that sparse couplings destroy it and whose §5.6 estimates the sparsity of the $\mathcal{N} = 4$ SYM supercharge as N^{-3} , the same scaling as the adjoint hypergraph density above; §5 below makes that heuristic precise for these models, although the window turns out to be governed by the symmetry class rather than the density. The one-flavour line at general odd q is the subject of Tierz [5], who computes BPS multiplicities for $\text{Tr}[\Psi^q]$ at $(q, N) = (5, 3), (5, 4), (5, 5), (7, 4)$ and finds a factorisation $Z_{\text{BPS}} = q^b x^{q_{\min}} (1+x)^N T(x)$. His $(1+x)^N$ is the factor the slot factorisation of §2 produces, his observation that the traceless diagonal modes make the cohomology a graded module over $\Lambda^\bullet(\mathbb{C}^{N-1})$ is the structure used there, with freeness left open on both sides away from the maximal weight, and his §6.1 measures the phenomenon this report is about. As a check on both codes, our machinery reproduces his $(5, 3)$ total exactly, $Z_{\text{BPS}} = 440$ with profile 20, 75, 125, 125, 75, 20.

2 The rank law

2.1 The maximal weight, and why only self-loops survive

The highest weight in the Fock space is found greedily, filling row 1 and emptying column 1, then row 2, and so on, giving the fully upper-triangular occupation with

$$\lambda_i^* = p(N+1-2i), \quad C_2^{\max} = \frac{1}{2}p(p+1) \sum_i (N+1-2i)^2 = \frac{p(p+1)N(N^2-1)}{6}. \quad (5)$$

N	window (N_Ψ)	W	BPS multiplets h^k
2	5, 6, 7	3	9, 18, 9
3	12, ..., 15	4	27, 81, 81, 27
4	22, ..., 26	5	81, 324, 486, 324, 81
5	35, ..., 40	6	243, 1215, 2430, 2430, 1215, 243

Table 1: Exact BPS multiplets of the maximal irrep at $p = q = 3$, as predicted by (6): $3^N \binom{N}{j}$ over $N + 1$ consecutive charges centred on $pN^2/2$, totalling 6^N . The factor $(1 + t)^N$ is the same one that appears in Chen’s single-matrix Z_{BPS} .

At $p = 1$ this is $N(N^2 - 1)/3$, Chen’s $C_2(r_*)$, so (5) is its multi-flavour generalisation. Shifting λ^* to a partition gives row lengths $p(2N - 2i)$, exactly p times Chen’s staircase with the same $N - 1$ non-empty rows; adding flavours stretches rows without producing new ones, a point we return to in §6.

At λ^* the off-diagonal sector is rigid, every upper mode occupied and every lower mode empty (Fig. 1a), so the weight space is an exterior algebra on the zero-weight modes, $(\Lambda V)_{\lambda^*} \cong \Lambda^\bullet(\mathcal{Z})$ with $\mathcal{Z} = \mathfrak{h} \otimes \mathbb{C}^p$ of dimension pr . Which terms of Q survive is best seen graphically. Read a mode Ψ_{ij}^a as a directed edge $i \rightarrow j$ of colour a . A term is a trace, so the q modes it creates form a closed walk, and Pauli exclusion lets it act only if every edge of that walk is empty. At the maximal weight the empty modes are precisely those with $i > j$, together with whatever diagonal flavours remain free, so every step of a creatable walk satisfies $i_t \geq i_{t+1}$: the walk may descend or stay level but never climb, since all uphill edges are occupied. Closing it forces $i_1 = \dots = i_q$. Only self-loops survive, meaning q diagonal modes of distinct flavours at a single site. We checked this exhaustively at $N = 3$, $p = 3$, where all 192 non-zero actions of Q on the maximal-weight space are of exactly this form.

2.2 Factorisation

Because every surviving term sits at one site, $Q = \sum_{i=1}^r Q_i$ with Q_i acting inside site i ’s flavour space, and since the q creation operators anticommute only the totally antisymmetric part of the coupling survives: $Q_i = \omega_{\text{sl}} \wedge -$ on $\Lambda^\bullet \mathbb{C}^p$ with $\omega_{\text{sl}} = \text{Alt}(C)$. The Q_i act on disjoint tensor factors of $\Lambda(\mathcal{Z}) = (\Lambda^\bullet \mathbb{C}^p)^{\otimes r}$, so Künneth applies and Poincaré polynomials multiply,

$$Z_{\text{max}}(t) = t^{k_0} (z_{\text{slot}}(t))^r, \quad z_{\text{slot}}(t) = \sum_j \dim H^j(\Lambda^\bullet \mathbb{C}^p, \omega_{\text{sl}} \wedge) t^j, \quad (6)$$

with $k_0 = p|\Delta^+|$ counting the frozen off-diagonal modes. Multiplying r copies of a polynomial adds their degree spans, so if z_{slot} is supported on w_s degrees then

$$W = r(w_s - 1) + 1. \quad (7)$$

This is the rank law. It is worth running once. At $p = q = 3$ and $N = 3$, the slot algebra $\Lambda^\bullet \mathbb{C}^3$ has dimensions 1, 3, 3, 1, and $\text{Alt}(C)$ spans the one-dimensional $\Lambda^3 \mathbb{C}^3$, so $\omega_{\text{sl}} \wedge$ is non-zero only as $\Lambda^0 \rightarrow \Lambda^3$. Hence $H^0 = 0$ by injectivity, $H^1 = H^2 = 3$ because the differential neither enters nor leaves, and $H^3 = 1 - 1 = 0$, giving $z_{\text{slot}} = 3t(1 + t)$. Cubing and restoring the prefactor yields $Z_{\text{max}} = 27t^{12} + 81t^{13} + 81t^{14} + 27t^{15}$, in exact agreement with the computed spectrum (Table 1).

2.3 SYK sits at $r = 1$

Setting $r = 1$ in (7) gives $W = w_s$, and $r = 1$ is exactly the $N = 1$ member, so the rank law already contains SYK. This is not a coincidence of notation: z_{slot} is by definition the cohomology of a generic q -form on $\mathbb{C}^p$, which is the SYK computation. The single formula therefore covers both ends of the family, and the requirement $W \leq q$ becomes a condition on the rank,

$$r(w_s - 1) \leq q - 1. \quad (8)$$

Table 2 collects every model we have measured. The width is $r + 1$ throughout, including SYK. That $u(2)$ and $\mathfrak{su}(3)$, different algebras of different N and different dimension, give the same W is a non-trivial check that rank rather than N or $\dim \mathfrak{g}$ is the governing variable.

3 The slot width

By (7) everything reduces to w_s , the number of degrees carrying cohomology for $\omega_{\text{sl}} \wedge$ on $\Lambda^\bullet \mathbb{C}^p$. This is the rank-one problem, which is SYK itself, and it can be solved rather than tabulated.

model ($p = 3, q = 3$)	rank r	W	$W \leq q?$	evidence
SYK = $u(1)$, any p	1	2	yes	all p from 4 to 12, matched against the matrix code
$u(2)$	2	3	yes	complete spectrum, all four irreps give 9, 18, 9
$\mathfrak{su}(3)$	2	3	yes	maximal weight
$u(3)$	3	4	no	complete Casimir tiers 48 down to 24, all 13 irreps
$\mathfrak{su}(4)$	3	4	no	maximal weight
$u(4)$	4	5	no	13 irreps down to $C_2 = 107$
$\mathfrak{su}(5)$	4	5	no	maximal weight
$u(5)$	5	6	no	maximal weight

Table 2: The rank law in measured form: $W = r + 1$ across every model computed, with SYK the rank-one case. Concentration survives exactly to rank 2.

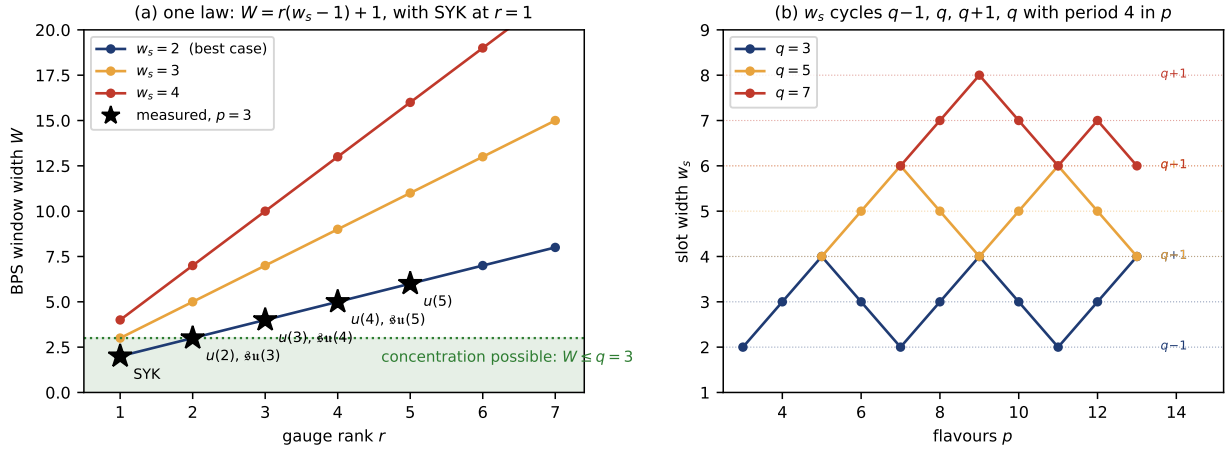


Figure 2: (a) The rank law with the measured models on it; SYK is not a separate case but the point $r = 1$. (b) The slot width for generic couplings, cycling through $q-1, q, q+1, q$ with period four in p as the bilinear form (15) alternates between symmetric and antisymmetric.

3.1 Each class saturates its index

Grade $\Lambda^\bullet \mathbb{C}^p$ by $j \bmod q$ and let

$$E_c = \sum_{j \equiv c \pmod{q}} (-1)^j \binom{p}{j} = \frac{1}{q} \sum_{m=0}^{q-1} \zeta^{-mc} (1 - \zeta^m)^p, \quad \zeta = e^{2\pi i/q}, \quad (9)$$

be the Witten index of class c , a topological invariant independent of the coupling. Since $\sum_c E_c = (1-1)^p = 0$ exactly one non-zero E_c is impossible, and since no factor $1 - \zeta^m$ vanishes they cannot all be zero, so at least two classes carry cohomology. This gives $w_s \geq 2$ and hence $W \geq r + 1$ at the maximal weight: concentration already requires $r \leq q - 1$, whatever the flavour content.

Computation says considerably more. For generic ω_{sl} every class with $E_c \neq 0$ carries cohomology in a *single* degree, of dimension exactly $|E_c|$. We find no exception in twenty-seven cases, checked class by class rather than in total: $q = 3$ with $p \leq 13$, $q = 5$ with $p \leq 14$, $q = 7$ with $p \leq 13$. Saturation of the index is therefore the same statement as R-charge concentration for the rank-one model, and (9) gives the BPS degeneracies in closed form. At most one class ever has $E_c = 0$, and it is either empty or carries two degrees, so

$$w_s = \begin{cases} q-1, & \text{the zero-index class is empty,} \\ q, & \text{no class has vanishing index,} \\ q+1, & \text{the zero-index class is occupied.} \end{cases} \quad (10)$$

The rest of this section decides which case occurs.

3.2 Adjoining a flavour is a mapping cone

Adjoin one mode e to $\mathbb{C}^p$. Every state either contains it or does not, so $\Lambda^\bullet \mathbb{C}^{p+1} = \Lambda^\bullet \mathbb{C}^p \oplus e \wedge \Lambda^\bullet \mathbb{C}^p$, and the form splits the same way as $\omega_{p+1} = \alpha + e \wedge \beta$ with $\alpha \in \Lambda^q \mathbb{C}^p$ and $\beta \in \Lambda^{q-1} \mathbb{C}^p$. Writing a general element as $x + e \wedge y$, using that β has even degree so passes e freely while α has odd degree so costs a sign, and using $e \wedge e = 0$,

$$\omega_{p+1} \wedge (x + e y) = \alpha x + e(\beta x - \alpha y), \quad \text{that is} \quad \omega_{p+1} \wedge = \begin{pmatrix} \alpha & 0 \\ \beta & -\alpha \end{pmatrix}. \quad (11)$$

The new complex is two copies of the old one glued by β . Were β to act trivially the copies would not interact and $Z_{p+1}(t) = (1+t)Z_p(t)$; every departure from that free doubling is caused by β , and conversely every factor of $1+t$ records a step at which β failed to act.

Since $q(q-1)$ is even, α and β commute, so β carries α -cocycles to α -cocycles and coboundaries to coboundaries, and descends to

$$\delta = \beta \wedge - : H_p^j \longrightarrow H_p^{j+q-1}. \quad (12)$$

The long exact sequence of (11) then gives

$$\dim H_{p+1}^m = \dim \text{coker}(\delta : H_p^{m-q} \rightarrow H_p^{m-1}) + \dim \ker(\delta : H_p^m \rightarrow H_p^{m+q-1}), \quad (13)$$

a new class being either something in the e copy that β failed to hit or something in the plain copy that β annihilates. We have checked that (13) reproduces every step from $p = 4$ to $p = 13$ at $q = 3$. Everything in it is fixed by the inductive hypothesis except the rank of δ .

3.3 Why δ usually cannot act

Because δ raises degree by $q-1$ it sends class c to class $c-1$, and by saturation the occupied degrees at level p form a run of w_s consecutive integers. So δ can be non-zero only between two occupied degrees exactly $q-1$ apart, and the number of such pairs is $\max(0, w_s - (q-1))$. The three cases of (10) therefore carry zero, one and two maps respectively, which settles two of them at once. When $w_s = q-1$ no δ exists, the doubling is free, and w_s rises to q . When $w_s = q+1$ there are two maps; in every case computed both are square of dimension one and both have full rank, and w_s falls back to q .

3.4 The remaining map is a bilinear form, and its symmetry decides

Everything thus turns on the single map present when $w_s = q$. Particle-hole conjugation gives $h^j = h^{p-j}$, so the occupied degrees are symmetric about $p/2$ and the outermost pair is j and $p-j$ with

$$j = \frac{1}{2}(p - q + 1). \quad (14)$$

Since $j + (q-1) = p-j$, the source and target of δ are Poincaré dual. The map is then not merely linear but a bilinear form on H^j ,

$$\langle a, b \rangle_\delta := [\beta \wedge a \wedge b]_{\Lambda^p}, \quad \langle b, a \rangle_\delta = (-1)^{j^2} \langle a, b \rangle_\delta = (-1)^j \langle a, b \rangle_\delta, \quad (15)$$

symmetric when j is even and antisymmetric when j is odd. The two behave in opposite ways. A generic symmetric form is non-degenerate, so δ has full rank, the two outer classes cancel against each other, and w_s drops to $q-1$. An antisymmetric form has even rank, so on the odd-dimensional space H^j it *cannot* be non-degenerate; a residue survives, the zero-index class becomes occupied, and w_s rises to $q+1$. Hence w_s cycles with period four in p ,

$$q-1 \longrightarrow q \longrightarrow q+1 \longrightarrow q \longrightarrow q-1 \longrightarrow \dots \quad (16)$$

turning wherever j is odd, that is at $p \equiv q+1 \pmod{4}$. This is the period-four pattern in the measured slot widths (Fig. 2b), now with a reason attached.

We built δ explicitly over $\mathbb{F}_P$, with every rank confirmed against a second prime, rather than inferring it from Betti numbers. The prediction holds in all nine available cases. At $q = 3$ the coranks are 1, 0, 3, 0, 1 for $p = 4, 6, 8, 10, 12$; at $q = 5$ they are 1, 0, 1, 0 for $p = 6, 8, 10, 12$; and in each series the degenerate entries are exactly those with j odd. The one feature not explained by (15) is the size of the defect: a generic antisymmetric form on an odd-dimensional space has corank one, which is what occurs except at $q = 3, p = 8$, where the corank is three. That case sits at $p = q^2$, and the coranks are powers of q , which suggests a q -adic mechanism of the same kind as the unexplained factor q^b in Tierz's factorisation.

3.5 Consequences for concentration

Since $w_s \geq q - 1$, the best available rank law is $W = r(q - 2) + 1$, and (8) becomes

$$r \leq \frac{q - 1}{q - 2}. \quad (17)$$

At $q = 3$ this permits rank 2; for every $q \geq 5$ it permits only rank 1, which is SYK. The cubic supercharge is thus the only degree that tolerates a gauge group at all, not by convention but because raising q steepens the rank law faster than it lifts the ceiling. The concentrating members of the family are therefore SYK at any q , together with $u(2)$ and $\mathfrak{su}(3)$ at $q = 3$.

One apparent loophole should be recorded and closed, because closing it sharpens what the rank law may be asked to do. The above assumes $p \geq q$. If $p < q$ then $\Lambda^q \mathbb{C}^p = 0$, the slot differential dies and $z_{\text{slot}} = (1 + t)^p$, so $w_s = p + 1$. At one flavour this gives $w_s = 2$ and hence $W = r + 1$ at the maximal weight for any q , which we confirm at $(q, N) = (3, 3), (3, 4), (5, 3), (5, 4), (7, 4)$, in every case with the binomial profile $\binom{N}{j}$. It is tempting to conclude that $q \geq N + 1$ then concentrates at every N . It does not: the window of the *total* cohomology exceeds the maximal-weight window once $q \geq 5$. At $(q, N) = (5, 3)$ the full Fock space carries $Z_{\text{BPS}} = 440$ BPS states over six sectors $R = 2, \dots, 7$ with profile 20, 75, 125, 125, 75, 20, against a maximal-weight window of four; the residues mod 5 are 2, 3, 4, 0, 1, 2, so $R = 2$ and $R = 7$ share a class and concentration fails. That total agrees exactly with Tierz [5].

Two things follow. The loophole is closed, and, more importantly, the irrep-independence of the window, which we verify at $q = 3$ through the Casimir descents cited in Table 2, is *false* at $q = 5$. The rank law is a statement about the maximal weight and we use it as such: at $q = 3$ the descents license reading it as a statement about the spectrum, while at larger q they do not, and since the total window is then wider than $r + 1$ the conclusion that raising q hurts is reinforced rather than weakened. A further caveat attaches to the whole maximal-weight analysis: the multiplicity 6^N of Table 1 is not macroscopic, so the complexes reached here are not yet the ones the concentration hypothesis is about.

4 Why rank: the Weyl group

The rank law says what happens. This section identifies what produces it.

4.1 The Cartan sector carries a Weyl invariant form

By §2 the maximal-weight complex is the Cartan sector: pr modes carrying the q -form $\omega|_{\mathcal{Z}}$. Gauge invariance does not merely restrict ω to traces. Restricted to the Cartan it says something sharper, namely that $\omega|_{\mathcal{Z}}$ is invariant under the normaliser $N(T)$, hence under the Weyl group $\mathcal{W} = N(T)/T$, which acts on $\mathfrak{h}$ and trivially on flavour. For $u(N)$ and $\mathfrak{su}(N)$ that group is S_N permuting the sites.

This is the whole obstruction, and it is visible by comparison. A generic, unconstrained q -form on $\mathbb{C}^n$ has bounded window: at $q = 3$ we measure $W = 3, 4, 3, 2$ according to $n \bmod 4$, for every n up to 13, never growing. That is the same computation as z_{slot} , and it is why SYK concentrates whenever $n \not\equiv 1 \bmod 4$. A Weyl invariant form does not behave this way.

The controlled experiment is to hold the Cartan size fixed and vary only the Weyl group. Taking $n = pr = 12$ modes throughout and letting (p, N) range over the factorisations of 12, so that $|\mathcal{W}| = |S_N|$ runs from 1 to 12! while nothing else changes, gives the following, each entry the narrowest window found over twenty random draws of a generic $\mathcal{W}$ -invariant form.

p	N	$ \mathcal{W} $	narrowest W over 20 draws
12	1	1	3 (this row is SYK)
6	2	2	3 (20 of 20 draws)
4	3	6	5 (20 of 20 draws)
3	4	24	5 (19 of 20; one draw gave 7)
2	6	720	11
1	12	12!	the invariant form vanishes, so $W = 13$

The window grows monotonically with the Weyl group at fixed mode count. Since lower semicontinuity makes generic couplings optimal, the third and fourth rows say that *no* choice of gauge invariant coupling achieves $W \leq 3$ at rank 3 or more, which is the no-go in its strongest measured form.

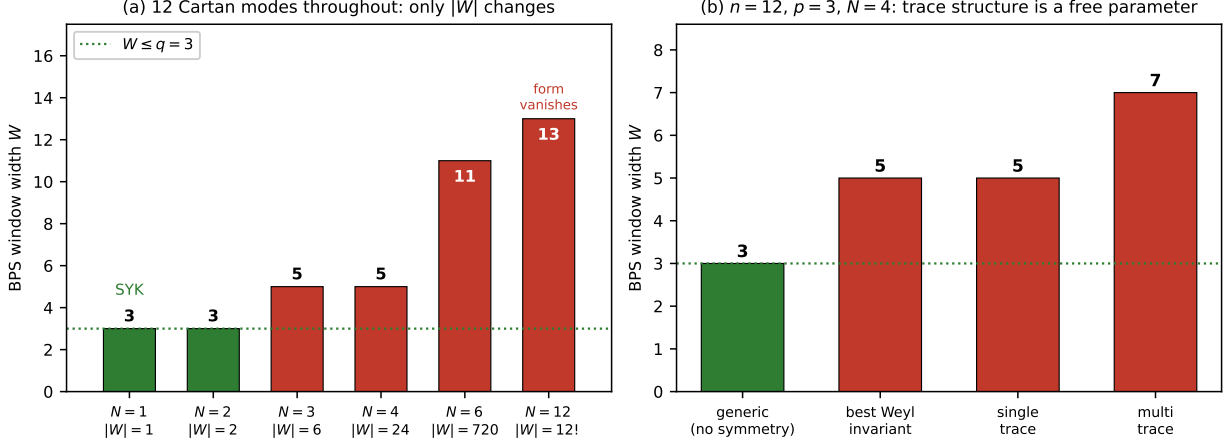


Figure 3: (a) Twelve Cartan modes throughout, with only the Weyl group changing. Concentration survives while $|\mathcal{W}| \leq 2$ and fails from $|\mathcal{W}| = 6$ on. The left-hand bar is SYK. (b) At $n = 12, p = 3, N = 4$: every Weyl invariant coupling, whatever its trace structure or density, sits well above the generic value.

4.2 The window sees the symmetry class and nothing finer

Membership in the Weyl invariant class is the whole of what the window responds to. The sharpest way to see this is to vary the trace structure, which changes the Cartan interaction drastically while staying inside the class. A single trace of q diagonal matrices localises, $\text{Tr}[H_1 \cdots H_q] = \sum_i \prod_t (H_t)_{ii}$, so a Cartan triangle needs q distinct flavours at the *same* site and the interaction is very sparse. Multi-trace terms are equally gauge invariant and do not localise: $\text{Tr}[\Psi^a] \text{Tr}[\Psi^b \Psi^c]$ restricted to the Cartan is $\sum_{i,j} \Psi_{ii}^a \Psi_{jj}^b \Psi_{jj}^c$, straddling two sites, and the triple trace straddles three. Since $\omega \wedge \omega = 0$ for any odd form, nilpotency is untouched, so these are legitimate models.

Their combinatorics is entirely different and their window is identical. Writing α_0 for the largest triangle-free set of Cartan modes, at $N = 4$ and $p = 3$ the single-trace model has four Cartan triangles, $\alpha_0 = 8$ and a bound of 5 from (19), which is attained. Switching on the double trace raises the triangle count to 100 and collapses α_0 to 5, making the bound vacuous; adding the triple trace takes the count to 124 with α_0 unchanged. The measured W is 5 in all three cases, and 7 if the single trace is dropped altogether.

Delocalising the Cartan interaction multiplies the triangle count by thirty and collapses α_0 from $2N$ to $N + 1$, and W does not move; only the edge multiplicities thin, from 27, 81, 81, 27 to 11, 81, 81, 11 at $N = 3$. Double-trace alone is wider still. This is what the Weyl picture predicts: all of these forms lie in the class Fig. 3a has already measured, and Fig. 3b shows each of them at or above the best value that class allows, well away from the generic one. Density and trace structure are free parameters within the class, and the window does not see them.

4.3 Most gauge groups are worse, and the unitary series is optimal

The next idea is to shop for a better gauge group, and Lie theory closes that down in one line. Suppose $-\text{id} \in \mathcal{W}$. It acts as -1 on every Cartan mode, hence as $(-1)^q = -1$ on a q -form with q odd, and q must be odd. Invariance then forces $\omega|_{\mathcal{Z}} = 0$, so Q vanishes on the maximal-weight complex, every state there is BPS, and $W = pr + 1$, the widest possible value.

The Weyl groups containing $-\text{id}$ are those of B_r, C_r, D_r with r even, G_2, F_4, E_7 and E_8 . Every orthogonal, symplectic and most exceptional gauge groups are therefore strictly worse than unitary. Only A_r, D_r with r odd, and E_6 escape, and among these $\mathfrak{su}(N)$ is what we have measured. The tempting option of a small Weyl group, $\mathfrak{su}(2)^r$ with $|\mathcal{W}| = 2^r$, is a trap for the same reason: $(\mathbb{Z}_2)^r$ contains $-\text{id}$, so its Cartan form vanishes identically. A Weyl group must be rich enough not to kill the form and small enough not to constrain it, and S_N with $N \leq 2$ is essentially the only solution.

5 A bound valid at every weight

The rank law is derived at the maximal weight, where the rigidity of the off-diagonal sector does the work. A weaker statement holds everywhere, by an elementary argument that is worth keeping because it is the only part of the analysis not confined to the top of the spectrum.

a state S is CRITICAL iff every triangle straddles (filled = occupied mode, open = empty)

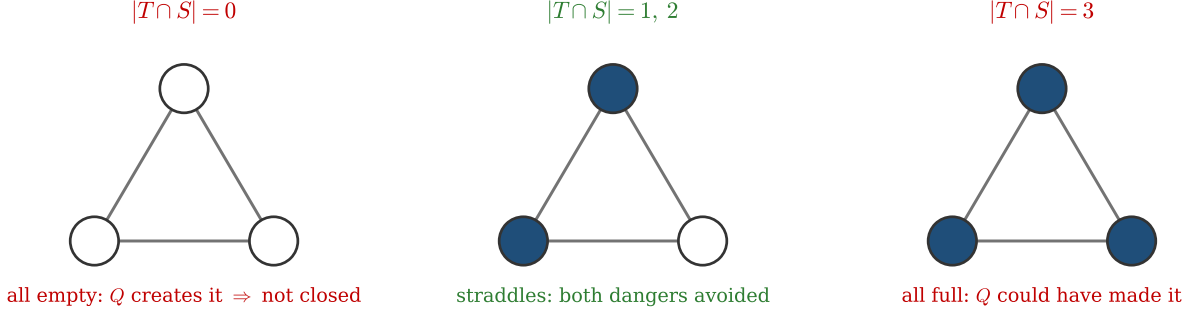


Figure 4: The three cases. If a triangle is entirely empty, Q can create it and S is not closed. If it is entirely occupied, S could have been produced by Q and the non-exactness argument fails. Only straddling triangles avoid both.

Recall that a triangle is the support of a non-zero term of ω . Call a set of modes S *critical* if every triangle meets it partially, $0 < |T \cap S| < 3$ for all T ; equivalently, neither S nor its complement contains a triangle. At $N = 3$, $p = q = 3$ there are 195 triangles on 27 modes out of $\binom{27}{3} = 2925$ possible triples, so the hypergraph is sparse, and criticality is easy to arrange.

The two inequalities do different jobs. The upper one makes m_S closed: for $\omega_T \wedge m_S$ to be non-zero all q modes of T must be absent from S , which criticality excludes, so every term is Pauli-blocked. The lower one makes m_S non-exact: since Q only ever adds a triangle, the coefficient of m_S in $\omega \wedge y$ is $\sum_{T \subseteq S} \pm c_T y_{S \setminus T}$, and criticality gives $T \not\subseteq S$ for every triangle, so that sum is empty for every y while m_S has coefficient 1 on itself. Each critical set thus gives a non-zero class in $H^{|S|}$, and distinct ones give independent classes.

Letting α be the largest triangle-free set and $\tau = n - \alpha$ the transversal number, S is critical exactly when both S and S^c are hitting sets, so critical sets live in $\tau \leq |S| \leq n - \tau$ and

$$W \geq n - 2\tau + 1 = 2\alpha - n + 1. \quad (18)$$

Wide windows are the signature of a hypergraph with large triangle-free sets.

Gauge invariance supplies such a set for free. Each term of Q is a singlet, so the weights of its q modes sum to zero, while inside $\mathfrak{n}^+ \otimes \mathbb{C}^p$ every mode carries a positive root and a non-empty sum of positive roots is never zero. No triangle fits in the positive-root subspace, for any reductive $\mathfrak{g}$ and any invariant coupling. With $|\Delta^+| = \frac{1}{2}(\dim \mathfrak{g} - r)$ this gives $\alpha \geq \frac{1}{2}(n - pr)$, and feeding it into (18) the root contribution cancels,

$$W \geq 2\alpha_{\text{cart}} - pr + 1, \quad (19)$$

where α_{cart} is the triangle-free number of the Cartan sector. The bound depends only on the rank and on the Cartan, which is the same conclusion §4 reaches by a different route. For single-trace $u(N)$ the Cartan piece gives $\alpha_{\text{cart}} = \min(p, q - 1)N$, whence

$$W \geq N(2\min(p, q - 1) - p) + 1, \quad (20)$$

which at $p = 1$ reads $W \geq N + 1$, reproducing Chen's single-matrix window combinatorially with no Casimir argument and no appeal to solvability. The explicit critical sets are $A_{\mathfrak{s}} = \{\text{all } p \text{ flavours of } E_{ij}, i < j\} \cup \{s_i \text{ flavours of } E_{ii}\}$ with $\max(0, p - q + 1) \leq s_i \leq \min(p, q - 1)$, triangle-free because the strictly upper part admits no closed loop and each site is one flavour short of a self-loop, with the complement triangle-free for the same reason.

The transversal number itself has a closed form, $\tau = N(N - 1)/2$ at $p = 1$ and $\tau = pN(N + 1)/2 - 2N$ for $p \geq 2$, agreeing with an exact integer program in all twelve cases tested (p from 1 to 4, N from 2 to 6). The resulting bound is attained for $p \leq 3$ and degrades at $p = 4$, where the family $A_{\mathfrak{s}}$ below is no longer optimal.

For SYK the same bound is vacuous, and instructively so. With generic couplings every q -subset is a triangle, so $\alpha_{\text{SYK}} = q - 1$ and (18) gives $W \geq 2(q - 1) - n + 1$, negative for $n \geq 2q - 1$. The transversal densities of Fig. 5 make the contrast concrete: $\tau/n = 0.83, 0.89, 0.93$ for SYK at $n = 12, 18, 27$, against $0.44, 0.46, 0.47$ for $u(3), u(4), u(5)$ at $p = 3$.

This bound is not tight in general, and the multi-trace couplings of §4 show the gap clearly: they drive it to a vacuous value while leaving W unchanged, exactly as expected once the window is known to depend on the

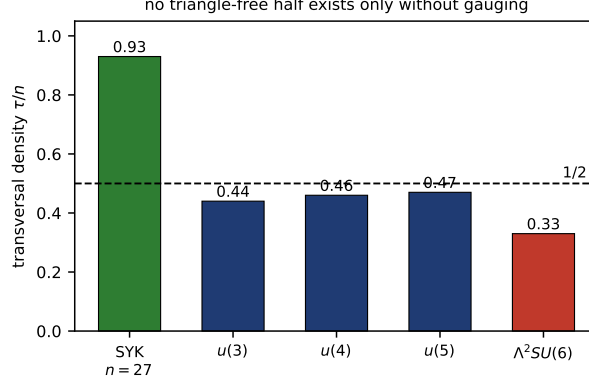


Figure 5: Transversal densities. SYK approaches 1, since no triangle-free set exceeds $q - 1$; gauge models sit near $\frac{1}{2}$, the positive-root half; two-index antisymmetric matter is worse still. This density governs (18); the window itself is set by the symmetry class (§4).

Fortuity: embedding $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ leaves the new site empty

(a) rank 3 (BPS state)

s_i	p	p
0	s_i	p
0	0	s_i

(b) rank 4 embedded: LIFTED

s_i	p	p	0
0	s_i	p	0
0	0	s_i	0
0	0	0	0

new site: self-loop
entirely EMPTY
 $\Rightarrow Q$ creates it

Figure 6: Embedding $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ as the upper-left block leaves every mode carrying the new index unoccupied. The self-loop at the new site is then an entirely empty triangle, so Q_{N+1} creates it and the state is lifted.

symmetry class rather than on the density. What (19) contributes that §2 does not is a statement holding at every weight.

6 Fortuity

For the single-matrix model the BPS states fill r_* , whose Young diagram is the staircase with $2N - 2i$ boxes in row i and exactly $N - 1$ rows. Going down in rank that diagram has too many rows to exist as a $U(M)$ irrep; going up it is no longer the maximal-Casimir representation. The states exist at one rank only: they are fortuitous rather than monotone. By §2 the diagram of λ^* has rows $p(2N - 2i)$, the same $N - 1$ rows, so the argument generalises verbatim to all p . The difficulty is that our BPS states are not confined to one irrep: at $N = 3$, $p = 3$ all thirteen irreps with $C_2 \geq 24$ carry them, each with window width 4. A statement about λ^* alone cannot settle fortuity here, so we give a local argument instead.

Let ι embed $\mathfrak{gl}(N) \hookrightarrow \mathfrak{gl}(N+1)$ as the upper-left block. It is worth being careful about what ι does and does not give, since it is *not* a chain map: ω_{N+1} contains terms carrying the new index that $\iota(\omega_N)$ does not, so $\omega_{N+1} \wedge \iota(m_S) \neq \iota(\omega_N \wedge m_S)$ and no map on cohomology is induced. This is the observation of Tierz [5], whose remedy is to compare ranks by the projection $\pi_{M \rightarrow N}$ deleting every monomial that uses an index above N , which *is* a chain map, and to call a class fortuitous when it lies outside the image of the projections from all larger ranks.

What the triangle language supplies is the sharp local reason the inclusion fails, and it fails maximally. Under the embedding every mode carrying the index $N + 1$ is unoccupied, so the self-loop at the new site, $T = \{\Psi_{N+1, N+1}^{a_1}, \dots, \Psi_{N+1, N+1}^{a_q}\}$ with q distinct flavours, satisfies $T \cap S = \emptyset$ for any embedded state S . A triangle disjoint from S is precisely one that Q can create, the Pauli blocking of §5 being absent, so $Q_{N+1} \iota(m_S) \neq 0$: the embedded representative is not merely unlifted, it is never closed. For $p < q$ the same holds for the sets A_s via the mixed loop $i \rightarrow N + 1 \rightarrow j \rightarrow i$ with $i < j$, since the two edges through $N + 1$ are new and hence empty while the return edge (j, i) is lower-triangular and hence empty. We verified this at $p = q = 3$, where all 9 BPS states of rank 2 and all 27 of rank 3 fail to be closed at the next rank, with the disjoint triangle exhibited in each case. So the obstruction is uniform: it removes every BPS state at once, which is why no stabilisation map exists rather than one that happens to vanish.

The two halves of this report rest on the same object, an empty triangle, and pull in opposite directions. Enlarging the gauge group manufactures a fresh site whose self-loop is necessarily empty, which is what blocks monotony and so produces fortuity; the same enlargement raises the rank, which by §2 widens the window and destroys concentration. The two properties one wants a fortuitous matrix model to have respond to the single available structural knob with opposite signs, which may be why the combination has proved hard to realise.

7 Status, and what is open

Established. The reduction (3); the index-loop restriction and the factorisation (6) with the rank law (7); the bound $w_s \geq 2$ and hence $W \geq r + 1$ at the maximal weight; the identification of SYK as the $r = 1$ member, verified against the matrix code for p from 4 to 12; the vanishing of $\omega|_{\mathcal{Z}}$ whenever $-\text{id} \in \mathcal{W}$, and the resulting classification of gauge algebras; the closed and non-exact properties of critical states, (18), the positive-root observation and (19) for any reductive $\mathfrak{g}$, with (20) for $u(N)$; the mapping-cone recursion (13) and, given saturation, the symmetry argument (15) that produces the cycle (16); and the failure of every embedded BPS state to be closed at the next rank.

Measured, not proved. Three loadbearing statements are numerical. Index saturation, on which §3 rests, is verified class by class in twenty-seven cases but not proved; it is the single most valuable gap, since with it the whole of §3 becomes a theorem. The growth of W with $|\mathcal{W}|$ at fixed mode count rests on five points and twenty coupling draws each. And the claim that the window is the same in every irrep, not just at the maximal weight, rests on the complete spectra at $(p, N) = (1, 3), (1, 4), (2, 3), (3, 2)$ and the Casimir descents at $(3, 3)$ and $(3, 4)$; it is false at $q = 5$. Two smaller inputs to §3 are also observed rather than derived: that at most one class has vanishing index, and that H^j is odd-dimensional in the antisymmetric case, without which (15) forces nothing. All cohomology is computed by exact integer linear algebra over $\mathbb{F}_P$ with every rank confirmed against a second prime, cross-checked against the refined Witten index, against exact diagonalisation at $N = 2$, and against Chen’s closed form at $p = 1$.

Open.

- **Index saturation.** A proof that a generic q -form saturates its index class by class would make §3 a theorem and, with it, R-charge concentration for the rank-one model. Ranks are lower semicontinuous, so one explicit witness per (p, q) suffices; the obvious candidate is not one, since the decomposable form $\sum_i e_{3i-2} e_{3i-1} e_{3i}$ factorises by Künneth into support $m + 1$ and is the worst rather than the best case, being exactly the maximal-weight structure of §2. The residual defect is smaller: a generic antisymmetric form on an odd-dimensional space has corank one, which is what we see except at $q = 3, p = 8$, where it is three. That case sits at $p = q^2$ and the coranks are powers of q , pointing at the same unexplained q -adic divisibility as the factor q^b in Tierz’s factorisation.
- **The Weyl growth, analytically.** A proof that a generic $\mathcal{W}$ -invariant q -form on $\mathbb{C}^p \otimes \mathfrak{h}$ has window growing with r would turn §4 from five measured points into the theorem the rank law deserves.
- **An upper bound on W .** Everything here bounds the window from below, or computes it at the maximal weight. The complementary statement, that the cohomology vanishes outside a band, equivalently that $\omega \wedge$ is injective in low degrees, is verified in six models and four complete spectra but not established. With it, the rank law would fix the window for every irrep and the no-go would become spectrum-wide.
- **$\mathfrak{su}(N)$ away from the top.** The set A_s is not triangle-free there, since the Cartan is $(N - 1)$ -dimensional

and spread across sites, so Cartan triangles need not localise. The true transversal numbers are $\tau = 12, 21, 36$ for $N = 3, 4, 5$ at $p = 3$, giving bounds $1, 4, 1$ against measured widths $3, 4, 5$: valid, tight only at $N = 4$, vacuous at odd N . The factorisation (6) also fails, the profiles $13, 81, 81, 13$ and $15, 258, 486, 258, 15$ being non-binomial, although the rank law still predicts the widths correctly.

- **Macroscopic index.** Concentration is conditioned on macroscopic index, and the maximal irrep has multiplicity $6^N = e^{1.8N}$, not $e^{O(N^2)}$. The descents reach $C_2 = 24$ at $N = 3$ with Fock multiplicity 110 080, the most macroscopic complex within reach, and the window does not close there. Reaching a genuinely macroscopic complex needs a sparse exact rank routine.
- **Chaos.** Nothing here bears on whether these models are chaotic. The three-flavour H is not a Casimir function, so a super-Schwarzian near-BPS spectrum remains possible without concentration.

References

- [1] Y. Chen, *Revisiting fortuity in matrix models*, 2025.
- [2] C. Chang, Y. Chen, W. Sia, S. Yang, *Fortuity in SYK models*, 2024.
- [3] C. Chang, Y. Lin, *Words to describe fortuitous states*, 2024.
- [4] W. Fu, D. Gaiotto, J. Maldacena, S. Sachdev, *Supersymmetric SYK models*, Phys. Rev. D **95** (2017) 026009.
- [5] M. Tierz, *BPS spectra of $\text{Tr}[\Psi^p]$ matrix models for odd p* , arXiv:2604.27164, 2026.
- [6] G. Turiaci, E. Witten, *$\mathcal{N} = 2$ JT supergravity and matrix models*, 2023.